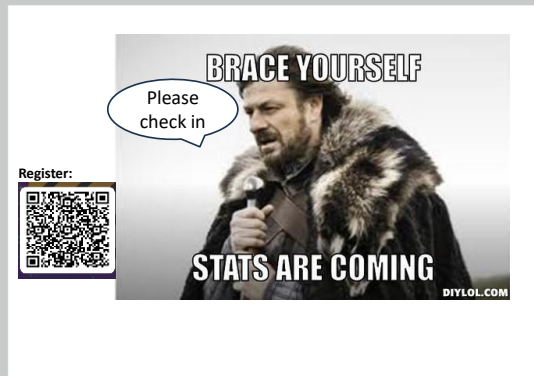


SEB Stats Workshop

Overview of Stats process

Materials and code:



Outline for today

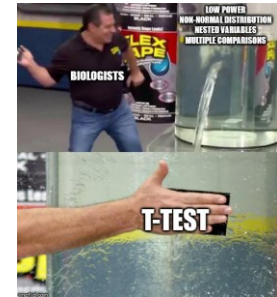
Morning

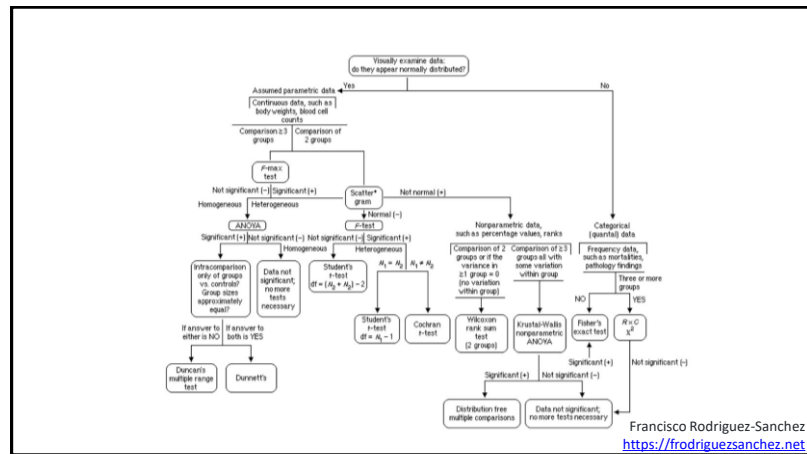
Lecture [1.5 hours, including Q&A]

- Stats basics
- Overview of Stats process

Worked example in R

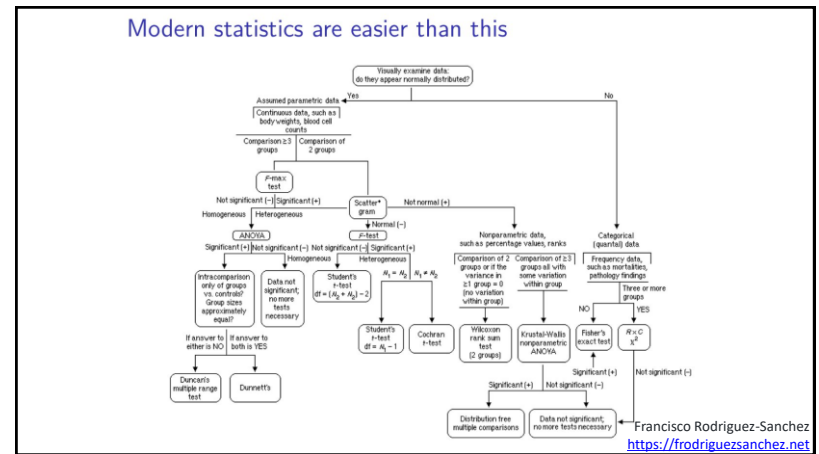
- [1.5 hours, including Q&A]
- GET CODE HERE:
- <https://github.com/hahnp13/Workshops.git>





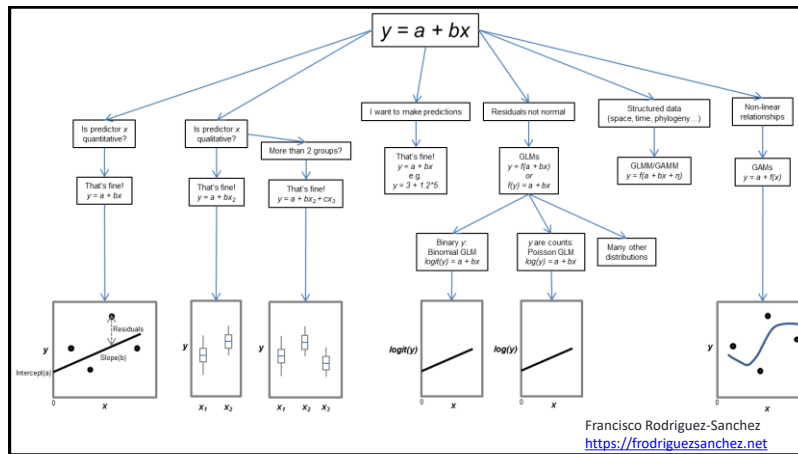
Francisco Rodriguez-Sanchez
<https://frodriguezsanchez.net>

4



Francisco Rodriguez-Sanchez
<https://frodriguezsanchez.net>

5



6



7

What is a probability distribution?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

8

What is a probability distribution?

1. A measure of how spread out the data values are from the mean
2. **A mathematical description of a random phenomenon that gives the probability of possible events**
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

9

What is a mean?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

10

What is a mean?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
- 3. A summary statistic that describes the center of a data set**
4. An estimate of the variability, or precision, of a sample mean

11

What is a standard deviation?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

12

What is a standard deviation?

- 1. A measure of how spread out the data values are from the mean**
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

13

What is a standard error?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

14

What is a standard error?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
- 4. An estimate of the variability, or precision, of a sample mean**

15

Questions ???

Materials and code:

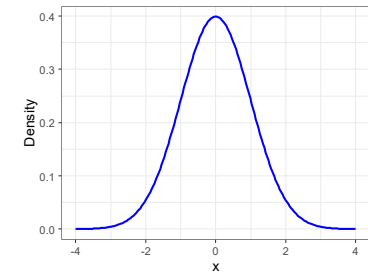


16

What is a probability distribution?

Normal (Gaussian) distribution

- Continuous numeric variables
 - $-\infty$ to ∞
 - Most points near the mean
 - Points symmetrically distributed around mean

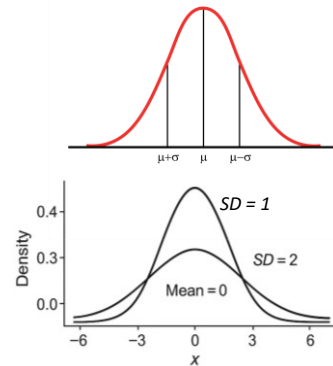


17

What is a probability distribution?

Normal (Gaussian) distribution

- Continuous numeric variables
 - $-\infty$ to ∞
 - Most points near the mean
 - Points symmetrically distributed around mean
- Parameters
 - μ = Mean
 - σ^2 = Variance
 - σ = Standard deviation



18

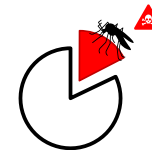
What processes generate data?

- Data are the result of natural (or biological) phenomena

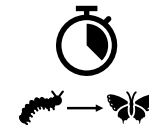
Counts



Proportion



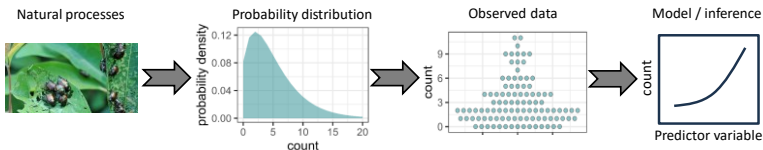
Time-to-Event



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What processes generate data?

- Data are the result of natural (or biological) phenomena
- Phenomena is mathematically described by a probability distribution
- Data we observe are a random subset of events produced by a phenomenon
- Used to model data in statistical analysis and make inferences



20

Steps for constructing and interpreting a GLMM

0. Data Exploration
1. Construct model
 - a. Fixed effects (predictor or explanatory variables)
 - b. Random effects (blocking variables)
 - c. Error distribution (response variable or dependent variable)
2. Assess residuals and overdispersion
3. Check random effects and/or distribution using AIC (not always needed)
4. Examine significance of terms in model
 1. Anova
 2. AIC model selection
5. Calculate treatment means and/or slopes
6. Construct contrasts
7. Calculate R²_m and R²_c
8. Make pretty figure

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Step 0: Data exploration

1. Formulate biological hypothesis
Carry out experiment
& collect data
- 2. Data Exploration**
3. Apply statistical model

Use to choose an appropriate model
and avoid assumption violations



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Step 0: Data exploration

1. Formulate biological hypothesis
Carry out experiment
& collect data
- 2. Data Exploration**
3. Apply statistical model

1. Understand design & variables
2. Explore data
 - a. Values & Outliers
 - b. Distribution & Variance
 - c. Collinearity
 - d. Relationships / Interactions
 - e. Independence / Clustering
3. Document decisions



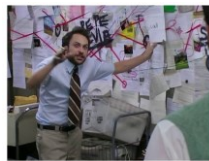
Michael A. Bishop and Christine 2015, 1, 14
DOI: 10.1101/029124/2015.04.04.088101.v1
A protocol for data exploration to avoid common statistical problems
Adam P. Best^{1,2}, Elena N. Ivan^{1,2} and Chris B. Bishop¹
¹Department of Biology, University of Cambridge, Cambridge, UK and ²Department of Biology and Evolutionary Biology and Center for Conservation Biology, University of Connecticut, Storrs, CT, USA

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Know your data and design

Get familiar with your dataset!

- What is the response?
What are predictors?
- Units, scales, coding
- Sampling design, repeated measures, groups



Explaining my datasheet to my colleagues



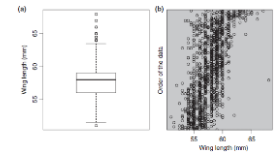
Toolbox

Summary statistics; frequency tables; generalized pairs plot

24

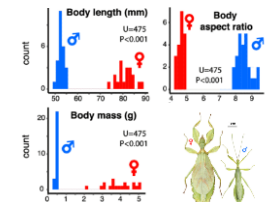
Explore Data - Values & Outliers

- Spot "impossible values" & data errors
- Outliers / fringeliers → Investigate



Toolbox

Histograms; box/violin/bean plots; dotplots; scatterplots



Investigate Subgroups

(e.g. season or sex) to see whether there are differences among subsets of the data.

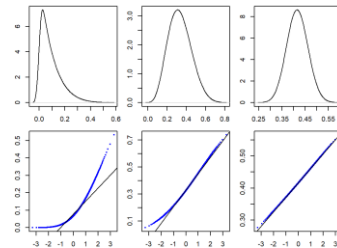
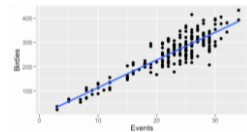


25

Explore Data - Distribution & Variance

Normality, skew, heteroscedasticity, ...

- Response: skew, zeros, heavy tails
- Predictors, floors/ceilings
- Plot against reference distribution
- **Use exploration to guide choices:** distribution, transformations



Toolbox

Histograms; density; (conditional)
line plots box/violin/MD; QQ plots

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Explore Data - Collinearity

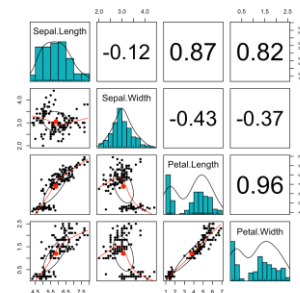
Collinearity happens when **predictors** are strongly correlated with each other

- Drop redundant variables,
- Combine them into an **index** or principal component
- Re-specify the model so each predictor has a clearer role (**record rationale!**)



Toolbox

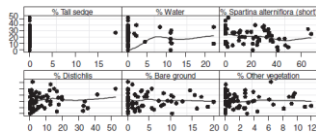
Correlation heatmap;
scatterplot matrix; VIF table



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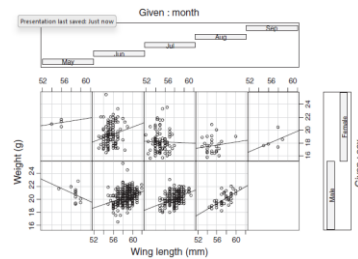
Explore Data – Relationships / Interactions

Relationships between response and each covariate



Toolbox
(Multi-panel) scatterplots;
conditional boxplots; coplots

Interactions between 3+ variables

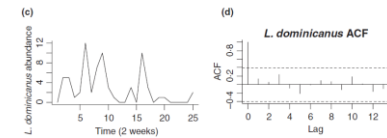


Explore Data - Independence / Clustering

- Grouping: subjects, sites, time points
- Plots by group / over time
- Need for random effects or correlation structures?



Toolbox
Spaghetti plots; grouped
box/dotplots; colored scatter;
maps



Good habits & Summary

Document your decisions

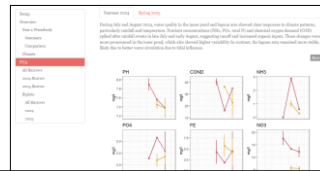
- Keep an “exploration log”
 - what you plotted, anomalies you saw, what you changed, and why
- Separate exploration vs confirmatory testing
- Share code, diagnostics, rationale



Toolbox
Rmarkdown, R/Jupyter notebooks;
saved plot grids; narrative logs

Summary

- Visualize first, model second
- Go through the checklist
- Use exploration to improve, not “game” statistics



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Step 1. Construct a statistical model to test your hypothesis

1. Construct a statistical model to test your hypothesis.

- Linear model: $y \sim x_1 + x_2 + x_1:x_2$
- Random effects?
- Distribution? Gaussian okay or generalized model?

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Step 1. Construct a statistical model to test your hypothesis

- Linear model:

$$y_i \sim \mu + \beta + e_i$$

Response variable mean Residual variation or error

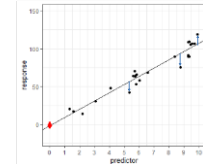
Parameter estimate to change the mean based on some variable (ie. slopes or intercepts)

32

Step 1. Construct a statistical model to test your hypothesis

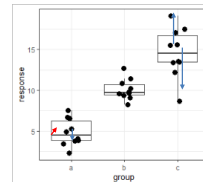
- “Linear regression”: $y_i \sim B_0 + B_1 X + e_i$

intercept slope resid



- “ANOVA” model: $y_{ij} \sim B_0 + \beta_j + e_{ij}$

intercept resid
Adjustment to intercept for each group



33

Questions ???



Materials and code:



Linear mixed models

- “ANOVA” model:

$$y_{ij} \sim B_0 + \beta_i + e_{ij}$$

- “RCBD”:

$$y_{ij} \sim B_0 + \beta_i + \alpha_j + e_{ij}$$

response

Predictor variable or fixed effects
(ie. different intercepts for
treatment means)

Random effect (ie. random intercepts for blocks)

Fixed vs. random effects?

Fixed effects

Random effects

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Fixed vs. random effects?

Fixed effects

- Usually the factor(s) of interest in study (or covariates)
- Same anywhere you do it
 - Values are meaningful and not exchangeable
- Represent all possible levels of a factor (of interest)
 - Same for continuous variables
- **Conclusions restricted to those levels (group effects)**

Random effects

40

Fixed vs. random effects?

Fixed effects

- Usually the factor(s) of interest in study (or covariates)
- Same anywhere you do it
 - Values are meaningful and not exchangeable
- Represent all possible levels of a factor (of interest)
 - Same for continuous variables
- **Conclusions restricted to those levels (group effects)**

Random effects

- Usually not the focus of study
- Levels (categories) could be exchanged/substituted (eg. Individuals or blocks)
- Represent a (random) sample of a larger set of levels
- **Can extrapolate conclusions to the larger sample (population level)**
- Different statistical properties

Benefits of blocking

- Blocks are formed so that treatments within blocks are as homogenous as possible
- Treatments randomly assigned within blocks
- Used to reduce variation among experimental units
 - Allows to account for variation from block-to-block
- Ensures interspersed among treatments
- Can account for pseudo-replication (when present)

Rules of thumb about random effects

- Sometimes it's a judgement call of whether something should be fixed or random
- Random effects should have at least 5 levels (and typically much more)
 - 5+ blocks, 5+ populations, 5+ clutches
- Each level of a random effect should have >3 observations
 - For most levels
 - Ok if a few levels have only a few observations
 - *Should be relatively balanced among levels* (but not perfectly)

Harrison et al. (2018), A brief introduction to mixed effects modelling and multi-model inference in ecology. PeerJ 6:e4794; DOI 10.7717/peerj.4794

Examples of blocks

- Chambers or tanks



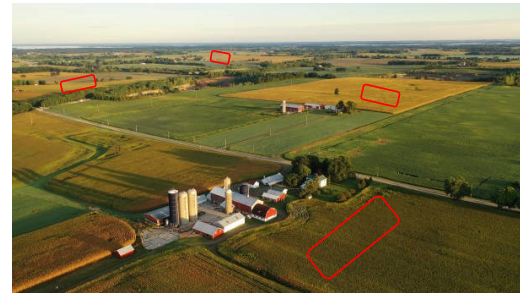
Examples of blocks

- Fields



Examples of blocks

- Farms

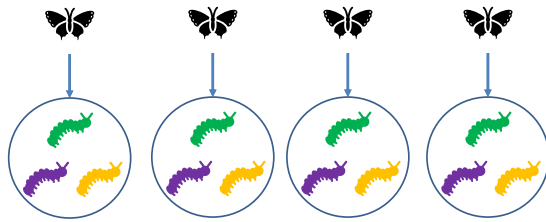


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Examples of blocks

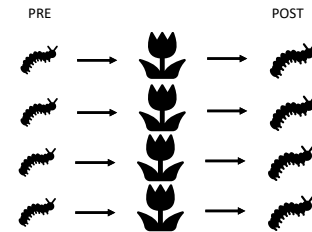
- Parent – siblings all similar to each other



48

Examples of blocks

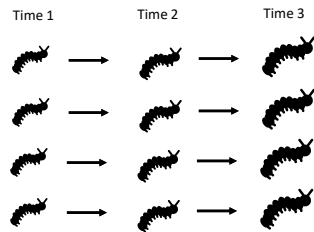
- Individuals/subjects
- Repeated measures



49

Examples of blocks

- Individuals/subjects
- Repeated measures



50



51

Which of the following would be characteristics typical of a **FIXED** effect in a statistical model? (select all that apply)

1. A factor of direct interest hypothesized to affect your response variable
2. A factor not of direct interest to the study which might have an unintended effect on your response variable
3. Values or levels are meaningful and the same anywhere you conducted the experiment
4. Values or levels could be exchanged or substituted
5. Values or levels represent a random subset of possible levels of the factor
6. Conclusions are restricted to the levels imposed
7. Conclusions can be extrapolated to the larger population

52

Which of the following would be characteristics typical of a **FIXED** effect in a statistical model? (select all that apply)

- 1. A factor of direct interest hypothesized to affect your response variable**
2. A factor not of direct interest to the study which might have an unintended effect on your response variable
- 3. Values or levels are meaningful and the same anywhere you conducted the experiment**
4. Values or levels could be exchanged or substituted
5. Values or levels represent a random subset of possible levels of the factor
- 6. Conclusions are restricted to the levels imposed**
7. Conclusions can be extrapolated to the larger population

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Step 1. Construct a statistical model to test your hypothesis

Table of R functions for constructing linear models

R function	SAS proc	Type of model	Type of test	Distribution
lm()	Proc glm Proc anova Proc reg	Linear model	Regression, ANOVA, ANCOVA	Gaussian
lmer()	Proc mixed	Linear mixed-effects models	Blocked or split-plot ANOVAs, nested designs, repeated measures.	Gaussian
glmer()	Proc glimmix	Generalized linear mixed-effects models	Blocked or split-plot ANOVAs, nested designs, repeated measures.	Poisson, Binomial
glmmTMB()	Proc glimmix	Any type of linear models, including generalized linear (mixed) models	Blocked or split-plot ANOVAs, nested designs, repeated measures.	Poisson, Binomial, Gamma, and many others

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Break



Materials and code:



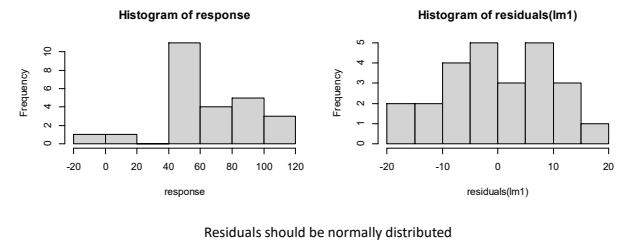
57

Step 2. Check model assumptions

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Assumptions of linear models

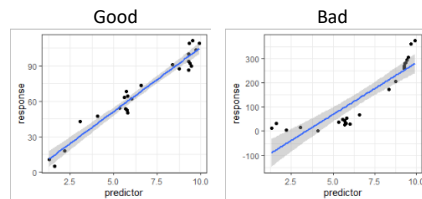
- **Response variable errors (ie. residuals) normally distributed**



59

Assumptions of linear models

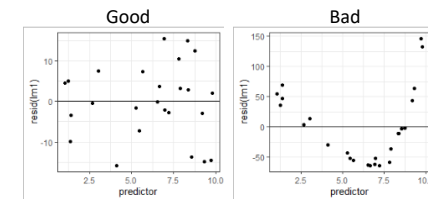
- Response variable errors (ie. residuals) normally distributed
- **Linearity**



60

Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- **Homoscedasticity**

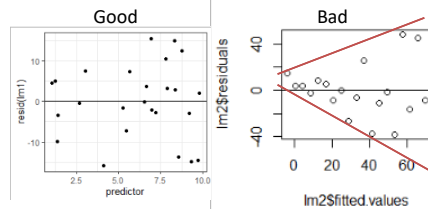


Residuals should be evenly scattered around zero when plotted against the original predictor

61

Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- **Homoscedasticity**

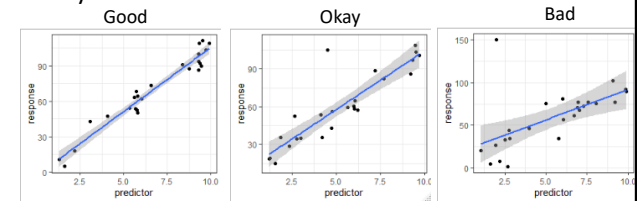


Residuals should be evenly scattered around zero when plotted against the original predictor

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Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- Homoscedasticity
- **Outliers**



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Often data (and residuals)
are not normally distributed

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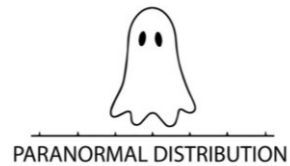
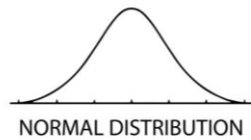
Why might we need to use non-normal distributions?

- Linear models are *reasonably* robust to violations of the assumptions
- Violations can affect
 - P-values
 - Standard errors
 - Parameter estimates
- Generalized models provide a better fit to the data
 - P-values
 - Standard errors
 - Parameter estimates
- Reviewers and editors expect it



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Questions ???



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What are “generalized” models?

- Linear models that allow for more flexible error distributions
- Parameters estimated via maximum likelihood (not least squares)
- Model components
 - Linear predictor formula ($y \sim x$)
 - Error distribution (usually exponential family)
 - Link function: links distribution to linear predictor formula
 - ie. Transforms error distribution to linear
- Generalized linear **mixed** model
 - Allows fixed and random effects

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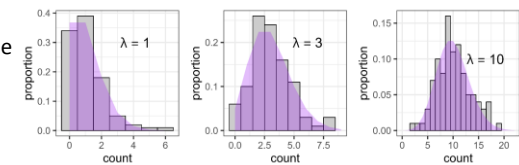
Generalized linear mixed models

- Linear model: $y_{ij} \sim \mu + \beta_i + \varepsilon_{ij}$
- Linear mixed model: $y_{ij} \sim \mu + \alpha_i + \beta_j + \varepsilon_{ij}$
- Generalized linear mixed model: $\log(y_{ij}) \sim \mu + \alpha_i + \beta_j + \varepsilon_{ij}$

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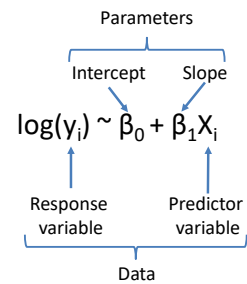
Poisson distribution - assumptions

- Discrete probability distribution
 - Integers: number of events (per unit of time or space; ie. rate)
 - Events occur randomly
 - Events are independent
- Parameters
 - λ = mean = variance
 - Link function = log

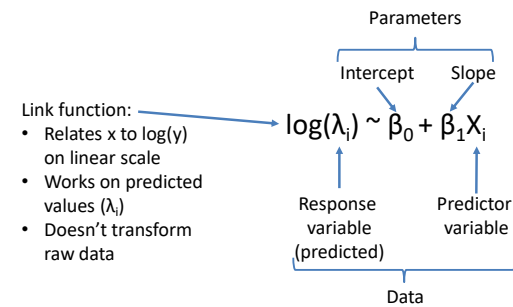


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Poisson linear model



Poisson linear model



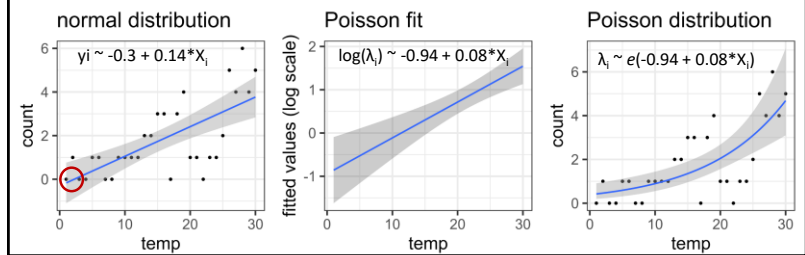
Poisson linear model

$$\begin{aligned}\log(\lambda_i) &\sim \beta_0 + \beta_1 X_i \\ &= \\ \lambda_i &\sim e^{\beta_0 + \beta_1 X_i}\end{aligned}$$

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Poisson linear model – example 1

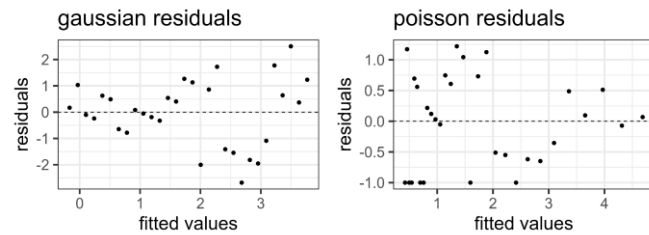
- Counts across temp (continuous, 1 to 30):
- $\log(\lambda_i) \sim \beta_0 + \beta_1 X_i$



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Poisson linear model – example 1

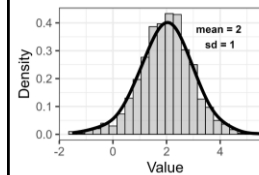
- Counts across temp (continuous, 1 to 30):
- $\log(\lambda_i) \sim \beta_0 + \beta_1 X_i$



Common distributions

Normal (gaussian)

- Continuous numeric
- Parameters
 - Mean
 - Variance (SD)
- Link = identity (none)



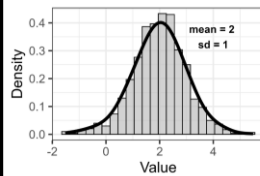
74

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Common distributions

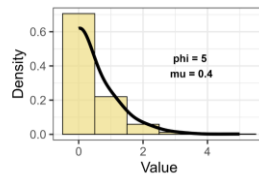
Normal (gaussian)

- Continuous numeric
- Parameters
 - Mean
 - Variance (SD)
- Link = identity (none)



Negative binomial

- Counts
- Parameters
 - μ (mean)
 - ϕ (precision)
 - Inverse of dispersion
- Link = log

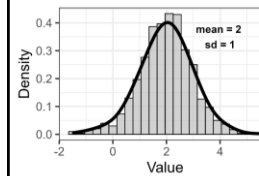


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Common distributions

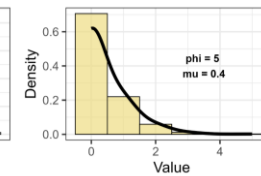
Normal (gaussian)

- Continuous numeric
- Parameters
 - Mean
 - Variance (SD)
- Link = identity (none)



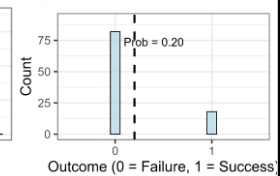
Negative binomial

- Counts
- Parameters
 - μ (mean)
 - ϕ (precision)
 - Inverse of dispersion
- Link = log



Binomial

- Proportions
 - **Binary** or ratio
- Parameters
 - Probability
 - # trials
- Link = logit

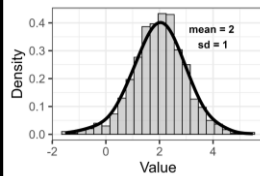


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Common distributions

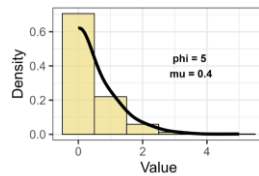
Normal (gaussian)

- Continuous numeric
- Parameters
 - Mean
 - Variance (SD)
- Link = identity (none)



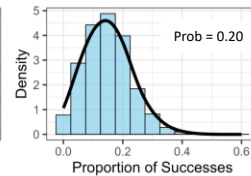
Negative binomial

- Counts
- Parameters
 - μ (mean)
 - ϕ (precision)
 - Inverse of dispersion
- Link = log



Binomial

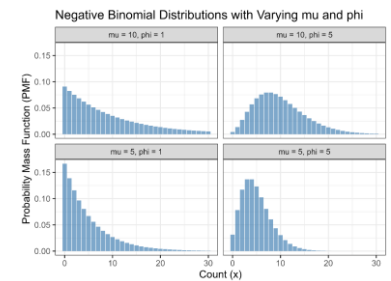
- Proportions
 - Binary or ratio
- Parameters
 - Probability
 - # trials
- Link = logit



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Negative binomial distribution - assumptions

- Discrete probability distribution
 - Integers: number of events
 - Events occur randomly (follow Poisson distribution)
 - Account for “extra” variation (beyond poisson)
- Parameters
 - μ = mean
 - ϕ = overdispersion parameter
 - lower values are more overdispersion
 - higher values converge to Poisson
 - $\text{Var} = \mu + \mu^2 / \phi$
 - Link function = log



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Common types of distributions

Distribution	Type of data	Overdispersion parameter?	Link
Gaussian	Continuous numeric variable	Yes	Identity
Log-normal	Continuous numeric variable	Yes	Identity
Poisson	Integers, counts	No	Natural log
Negative binomial	Integers, counts	Yes	Natural log
Binomial	Success/failure or presence/absence (binary), ratio	No	Logit, cloglog
Beta-binomial	Success/failure, presence/absence (binary), ratio	Yes	Logit, cloglog
Beta	Proportion (when the denominator is not known)	Yes	Logit
ordbeta	Proportion (when the denominator is not known), can have zero's	Yes	Logit
Gamma	Positive continuous variable	Yes	Inverse, log
ziGamma	Positive continuous variable, can have zero's	Yes	log

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What distribution best matches the data?

Description of data

1. Concentrations of chemicals in the soil, most values are low (~ 0.1 ug/L) but some are high (5-10 ug/L)
2. Number of infected and uninfected aphids on plants
3. Percent damage on leaves caused by herbivore or pathogens
4. Number of thrips on a plants

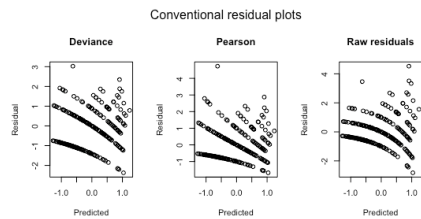
Distribution

- A. Normal
- B. Log-normal
- C. Poisson
- D. Negative binomial
- E. Binomial
- F. Beta
- G. Gamma

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Step 2. Check GLM assumptions

- “Conventional” residuals don’t work well for non-normal data



This figure shows residuals fit for count data assuming a Poisson distribution. The model (not shown) fits the data well, although the residuals look wonky.

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Step 2. Check GLM assumptions

- “Regular” residuals often don’t work well for non-normal data
- Simulated residuals offer a great diagnostic tool

DHARMA: residual diagnostics for hierarchical (multi-level/mixed) regression models

Florian Hartig, Theoretical Ecology, University of Regensburg [website](#)
2021-03-27

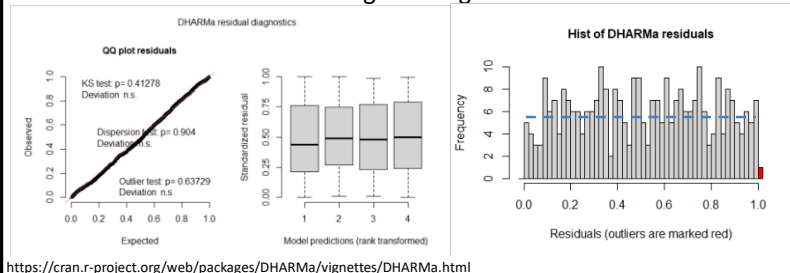
DHARMA stands for “Diagnostics for Hierarchical Regression Models”

<https://cran.r-project.org/web/packages/DHARMA/vignettes/DHARMA.html>

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Step 2. Check GLM assumptions

- “Regular” residuals often don’t work well for non-normal data
- Simulated residuals offer a great diagnostic tool



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Reasons for deviations of residuals

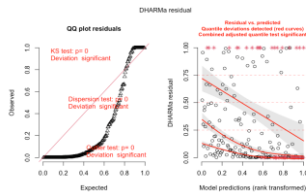
- Mis-specified model
 - Incorrect distribution (or potentially link function)
 - Missing fixed effect
 - Include additional covariates, if available
 - Check outliers in covariates or influential data points
 - Include quadratic term
 - Incorrect random-effect structure
 - Missing block or nestedness?
 - Incorrect order of nesting?
- **Overdispersion** (return to this later)
- Zero-inflation

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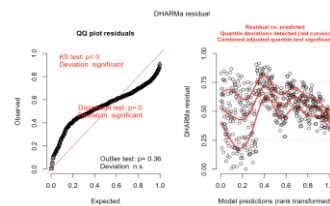
Step-by-step guide to analysis (post-data collection)

2. Check model assumptions

Bad residuals (overdispersion)



Bad residuals (wrong dist., overfitting, etc.)



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Step 3. Check random effects and model components

Investigate model with summary()

- Random effects
 - Examine variance components
- Fixed effects
 - Check fixed-effect signs, magnitudes, and uncertainty
 - Consider whether all predictors and interactions remain justified
- Fit metrics (e.g., AIC)
 - Compare a set of plausible models

Check random effects

- Not always necessary
 - Not all models have/need random effects and non-normal distributions
- Very small/zero variance components suggest simplification
 - Singularity can be worth noting (very small VC is *almost* singular)

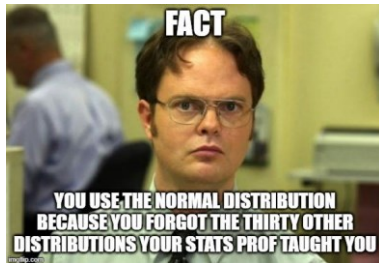


AIC for model comparison

- smaller AIC = better fit–complexity trade-off

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Questions ???



Materials and code:



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Step 4. Check significance of terms in model

Where do p-values come from?

- Estimate parameter values (i.e., slopes or intercepts)
- Calculate a test-statistic
- Use a distribution to find your p-value

Alternative method is AIC model selection

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Methods to estimate parameters

Least squares

- Minimizes the sums of squares
- $\sum (y_i - \bar{y})^2$

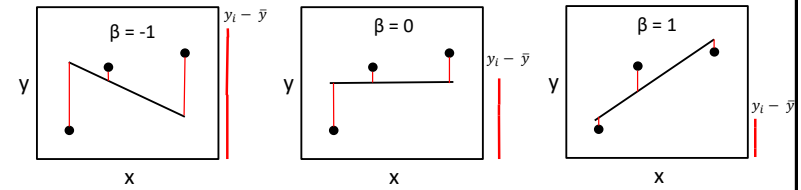
Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$

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Least squares estimation

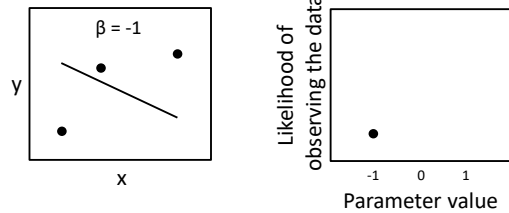
- Minimizes the sums of squares
- $\sum (y_i - \bar{y})^2$
- β = slope; Fit represents $\bar{y}$



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Maximum likelihood estimation

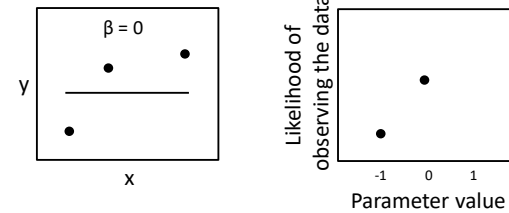
- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$: How likely is β given the observed data (y)
- β = slope



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Maximum likelihood estimation

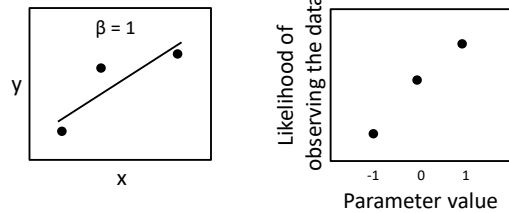
- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$: How likely (probable) is β given the observed data (y)
- β = slope



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Maximum likelihood estimation

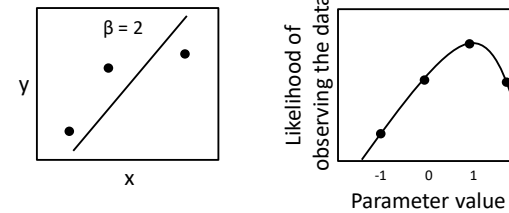
- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$: How likely (probable) is β given the observed data (y)
- β = slope



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Maximum likelihood estimation

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$: How likely (probable) is β given the observed data (y)
- β = slope

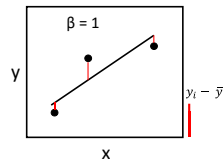


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Methods to estimate parameters

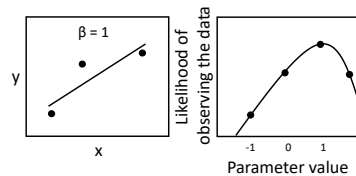
Least squares

- Minimizes the sums of squares
- $\sum (y_i - \bar{y})^2$
- β = slope or intercept adjust
- Only works for normal data



Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$
- β = slope or intercept adjust
- Can be generalized to any distribution

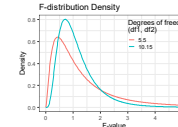


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Methods to estimate parameters

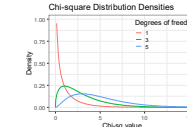
Least squares

- Minimizes the sums of squares
- $\sum (y_i - \bar{y})^2$
- β = slope or intercept adjust
- Only works for normal data
- F-value for significance test
 - $F = MS_B / MS_W$



Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$
- β = slope or intercept adjust
- Can be generalized to any distribution**
- χ^2 for significance test
 - $\chi^2 = (\text{effect size} / \text{SE})^2$



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After break

- Worked example
- Overview Steps 1-4
- **Step 5: Calculate group means**
- **Step 6: Conduct contrasts**
- **Step 7: Calculate R²**
- Step 8: Visualize data (if time permits)

Materials and code:



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Materials and code:



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